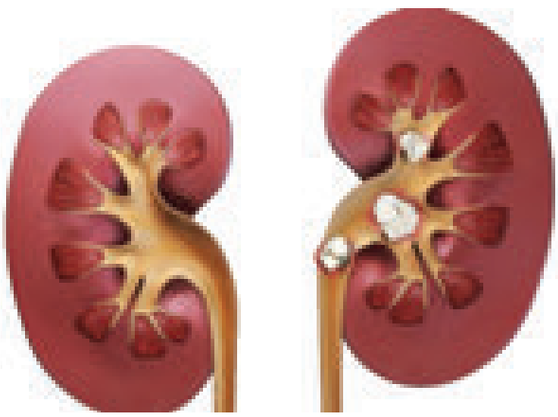
### Incidence

It is associated with urinary tract infection. Commonly seen between the age of 20 and 55yrs.

## Types of kidney stones

- **Calcium stones:** Are the most common type made of calcium and oxalate.35-40%
- **Cystine stones:** Are very rare 1-2%
- **Uric acid stones:** Formed when urine is too acidic.5-8%
- **Struvite stones:** Can happen when there is urinary tract infections.10-15%
- **Upper urinary tract stones:** That involve the renal pelvis and extend into at least 2 calyces are classified as *staghorn calculi*



## Causes

- Genetic factors-- family history of kidney stones
- Previous history of kidney stone.
- Certain medications
- Infection
- Urinary stasis
- Hyper Calcemia
- Hypercalcuria
- Diets that are high in protein and sodium but low in calcium
- Sedentary lifestyle, obesity, pressure, immobility
- Dehydration
- Warm climate

## Signs and Symptoms

- Severe pain and is called renal colic.
- Flank pain
- Haematuria
- Obstruction
- Infection
- Edema
- Pyuria
- Nausea and vomiting
- Fever with chills
- Frequent urination
- Urinating small amounts of urine
- Increased urge to urinate

## Diagnosis

- Health history
- Physical examination.
- 24 hour urine test
- Urography.
- Blood tests for calcium, phosphorus, uric acid, electrolytes, urea and nitrogen
- Urinalysis to check for crystals, bacteria, blood, and white cells
- Examination of passed stones to determine the type
- KUB X-rays
- Intravenous Pyelogram (IVP)
- Ultrasound of the kidney
- MRI of the abdomen and kidneys
- Abdominal CT scan

## Management

### Medical management

- Antispasmodic drug-relieves colic pain
- Antibiotics-prevents infection
- Narcotics-relieves pain

### Surgical management

- Cystoscopy: Scopic removal of stones



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